

DYNA DOXYCYCLINE CAPSULE 100MG

MAL19984405AZ

DESCRIPTION:

CAPSULE

Capsule Colour : Green/Green

Capsule Size : # 2

Powder Colour : Yellow

CONTENT:

Each capsule contains: Doxycycline HCl 115 mg equivalent to Doxycycline 100 mg

PHARMACODYNAMICS:

Doxycycline HCl has a broad spectrum of antimicrobial activity. It interferes with bacterial protein synthesis by reversibly binding to bacterial 30S ribosomal sub-unit, thus preventing the binding of aminoacyl transfer RNA to the messenger RNA ribosome complex. They are usually active against mycoplasmas, rickettsias, and spirochaetes, and also against many aerobic and anaerobic Gram positive and Gram negative pathogenic bacteria, and some protozoa. Although the microbial activity and mode of action of Doxycycline HCl is similar to Tetracycline, it is more effective than Tetracycline against many species.

PHARMACOKINETICS:

Doxycycline HCl is readily absorbed from the gastrointestinal tract and absorption is not significantly affected by the presence of food in the stomach or duodenum. Mean peak plasma concentrations of 2.6 µg per ml have been reported 2 hours after a 200 mg dose by mouth, falling to 1.45 µg per ml at 24 hours. Following repeated doses of 100 mg daily, concentrations of about 2 µg per ml are maintained. Similar values are reported after intravenous infusions. From 80 to 95% of Doxycycline in the circulation is reported to be bound to plasma proteins. Its biological half-life varies from about 15 to 22 hours. Doxycycline is more lipid-soluble than Tetracycline. It is widely distributed in body tissues and fluids but only low concentrations are achieved in cerebrospinal fluid. Doxycycline is slowly excreted in the urine and it is also excreted in the faeces following chelation in the intestines. There is also some inactivation in the liver. It is stated not to accumulate significantly in patients with renal impairment although excretion in the urine is reduced; increased amounts of Doxycycline are excreted in the faeces in these patients. Nevertheless there have been reports of some accumulation in renal failure. Removal of Doxycycline by haemodialysis is insignificant.

INDICATION:

Doxycycline has been found clinically effective in the treatment a variety of infections caused by susceptible strains of Gram-positive and Gram-negative bacteria and certain other micro-organisms.

Respiratory tract infections: Pneumonia and other lower respiratory tract infections due to susceptible strains of *Streptococcus pneumoniae*, *Haemophilus influenzae*, *Klebsiella pneumoniae*, *Mycoplasma pneumoniae* and other organisms.

Treatment of chronic bronchitis and sinusitis.

Urinary tract infections: Infections caused by susceptible strains of the *Klebsiella* species, *Enterobacter* species, *Escherichia coli*, *Streptococcus faecalis* and other organisms.

Sexually transmitted diseases: Infections due to *Chlamydia trachomatis* including uncomplicated urethral, endocervical or rectal infections.

Non-gonococcal urethritis caused by *Ureaplasma urealyticum* (T-mycoplasma).

Doxycycline is also indicated in chancroid and infections due to *Calymmatobacterium granulomatis*. Doxycycline is an alternative drug in the treatment of gonorrhoea and syphilis.

Dermatological infections: Acne vulgaris when antibiotic therapy is considered necessary.

Doxycycline is indicated for prophylaxis in the following conditions: 'Scrub Typhus', 'Travellers' diarrhoea' (enterotoxigenic *Escherichia coli*), leptospirosis.

RECOMMENDED DOSE:

Adults and children over 8 years weighing 45 kg or more: 100 mg at 12-hour intervals for two doses, followed by 100 mg once daily.

Children weighing less than 45 kg over 8 years of age or under special circumstances if younger: 2.2 mg/kg of body weight daily at 12-hours the first day then 2.2 to 4.4 mg per kg of body-weight once a day or 1.1 to 2.2 mg per kg of body-weight every 12 hours.

In severe infections in both adults and children, the initial daily dose may be given every 12 hours for as long as necessary. Therapy should be continued for at least 24 to 48 hours after signs and symptoms have subsided. For streptococcal infections, therapy should be continued for at least 10 days.

Nongonococcal urethritis caused by *Chlamydia trachomatis* or *Ureaplasma urealyticum*, and Uncomplicated urethral, endocervical, or rectal infection caused by *Chlamydia trachomatis*: 100 mg two times a day for at least seven days.

Syphilis (primary and secondary): 300 mg a day in divided doses for at least 10 days.

Traveler's diarrhea: 200 mg on the first day of the travel, followed by 100mg daily throughout the stay in the area.

Prevention of scrub Typhus: 200 mg as a single oral dose.

Prevention of Leptospirosis: 200 mg orally on a weekly basis throughout the stay in the area and 200mg at the completion of the trip.

Treatment of Leptospirosis: 100 mg orally twice daily for 7 days.

Acne Vulgaris: 50 - 100 mg daily for up to 12 weeks.

ROUTE OF ADMINISTRATION:

FOR ORAL USE ONLY

CONTRAINDICATIONS:

Doxycycline is contraindicated in pregnancy, infancy and childhood to the age of 8 years as it causes tooth discolouration.

WARNINGS AND PRECAUTIONS:

It should be used with caution in patients with hepatic dysfunction. Absorption from oral dose is impaired by antacids, mineral supplements such as iron, calcium or magnesium when taken concomitantly. Doxycycline may cause oesophagitis or oesophageal ulceration and tablets or capsules should therefore be given with at least 100 ml of water while the patient is in an upright sitting position.

Some patients with spirochete infections may experience a Jarisch-Herxheimer reaction shortly after doxycycline treatment is started. Patients should be reassured that this is a usually self-limiting consequence of antibiotic treatment of spirochete infections.

DRUG INTERACTION:

Combinations containing any of the following medications may interact with this medication: Antacids, Calcium supplements, Iron supplements, Magnesium-containing laxatives, Contraceptives, Penicillins and Barbiturates. Absorption from oral dose is not significantly affected by food or milk.

Carbamazepine or phenytoin: Concurrent use with Doxycycline may result in decreased Doxycycline serum concentrations due to induction of microsomal enzyme activity; adjustment of Doxycycline dosage or substitution of another Tetracycline may be necessary.

Methoxyflurane: Concurrent use with Tetracyclines may increase the potential for seprotoxicity.

Anticoagulant: Anticoagulant effect possibly enhanced by Tetracyclines.

PREGNANCY AND LACTATION:

Doxycycline is contraindicated in pregnancy. The risks associated with use during pregnancy are believed to be predominately associated with the effects on teeth and interference with and impairment of skeletal development in the unborn child. There is also a risk of hepatotoxicity in the mother. Tetracyclines are excreted in breast milk and are therefore contraindicated in nursing mothers. Results of animal studies indicate that Tetracyclines cross the placenta, are found in foetal tissues and can have toxic effects on the developing foetus (often related to retardation of skeletal development). Evidence of embryotoxicity has also been noted in animals treated early in pregnancy.

SIDE EFFECTS:

Discolouration of infant's or children's teeth. Cramps or burning of the stomach. Darkened or discoloured tongue due to fungal overgrowth. Hypertrophy of the papilla. Diarrhoea. Increased sensitivity to sunlight itching of the rectal or genital areas or sore mouth or tongue due to fungal overgrowth. Gastrointestinal disturbances including nausea and vomiting may occur.

Hypersensitivity reactions: urticaria, angioneurotic oedema, anaphylaxis, anaphylactoid purpura, pericarditis and exacerbation of system lupus erythematosus.

Immune system disorders: Frequency not known. Jarisch-Herxheimer reaction (see Section Warnings and Precautions).

Skin and subcutaneous tissue disorders: Frequency 'rare': Fixed eruption.

SYMPTOMS AND TREATMENT OF OVERDOSE:

Overdosage may cause GIT disturbances like nausea, vomiting, diarrhoea. In the treatment of overdosage, most patients required only supportive and symptomatic therapy. Discontinue the drug and treat GIT disturbances by giving milk or Charcoal by mouth to reduce their absorption. Dialysis does not alter serum half-life and thus would not be of benefit in treating cases of overdosage.

PACKING/PACK SIZE(S):

Blister pack of 1x10's, 10x10's, 50x10's and 100x10's.

**JAUHI UBAT DARIPADA KANAK-KANAK
KEEP OUT OF REACH OF CHILDREN**

MANUFACTURER/PRODUCT REGISTRATION HOLDER:

DYNAPHARM (M) SDN. BHD. 198001011897 (65683-V)

2497, Mk. 1, Lorong Perusahaan Baru 5,

Kawasan Perusahaan Perai 3,

13600 Perai, Pulau Pinang, MALAYSIA

P11PC D016-02
Latest Revision: Oct 2025